

Technology stack

Orchestration

Python 3.11 · FastAPI 0.111
LangGraph 0.1.5 · langchain-core
uvicorn · httpx · asyncio

LLM serving

HuggingFace router /v1
Llama 3.1 8B Instruct
OpenAI-compatible chat completions
tenacity retry · cold-start aware

Embeddings

BAAI/bge-large-en-v1.5
HF inference endpoint (CPU)
1024-dim · cosine similarity

Vector databases

Qdrant 1.9.2 (Docker)
vector_index_chunks collection
clara_memory_index collection

Search · Graph · Storage

Elasticsearch 8.13 · BM25
Neo4j 5.19 community · APOC
PostgreSQL 16 · asyncpg
Redis 7.2 · redis[asyncio]

Compression · Monitoring · Frontend

CLaRa via Llama 3.1 chat API
Prometheus + Grafana · /metrics
HTML/CSS/JS UI · FastAPI StaticFiles

Control & data flow

Flow 1 — user query arrives at POST /query
orchestrator/main.py → run_pipeline()

Flow 2/3 — Redis cache check
exact SHA-256 → semantic cosine → miss

Flow 4 — classify → route
LLM classifies to one of 5 query types

Flows 5a–5e — retrieval routing
Hybrid / GraphRAG / HopRAG / SG-RAG / Self

Flow 7 — all retrievers → compression
CLaRa memory runs concurrently via gather()

Flow 8 — compress → context assembly
skip if <200 words · 4x ratio target

Flow 9/10 — LLM generation
Llama 3.1 8B · /v1/chat/completions

Flow 11 — answer → cache store
exact + semantic written concurrently

Flow 13 — telemetry collection
Prometheus counters · latency histograms

Flow 14 — ingestion: chunk → embed → BM25 → KG → CLaRa → r